Computer Vision, Assignment 3
Epipolar Geometry

1 Instructions

The data for the assignments is available from the course page:

`https://canvas.education.lu.se/courses/39473`

Deadline:
Hand in on time via Canvas by study week 6, 1/3 at midnight. You must attempt all mandatory exercises (no blank answers). Extensions are only given in exceptional cases after emailing the lecturer in advance.

Report format:
Submit one PDF with complete solutions. Use LaTeX for typesetting the report. Handwritten solutions to some exercises are allowed if scanned into the PDF and clearly structured and readable.

Code submission:
Submit all code in a single zip file, including any data. Code must be well commented and directly runnable (no manual loading of data, setting parameters, etc.).

Help and collaboration:
You may work together and ask questions in Q&A sessions or by emailing the lecturer, but the report must be your own work. You may not use AI tools (e.g. ChatGPT or other LLMs) to directly construct your solutions. You must understand and be able to explain **everything** you submit, and you may be asked to clarify parts of your work if there is any doubt about authorship.

Corrections:
If any part of your submission is incorrect, you may submit corrections. You are allowed a maximum of two correction submissions.

Bonus:
The OPTIONAL exercises are not required to pass but give bonus points on the take-home exam if solved well. You can earn up to 0.25 points per assignment.

2 The Fundamental Matrix

Exercise 1. Let $P_1 = [I\ 0]$ and

$$P_2 = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 2 & 0 & 2 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \tag{1}$$

a) Compute the fundamental matrix.

b) Suppose that the point $x = (1, 1)$ is the projection of a 3D-point $\mathbf{X}$ into P_1. Compute the epipolar line in the second image generated from x.

c) Which of the points $(2, 0)$,$(2, 1)$ and $(4, 2)$ could be a projection of the same point $\mathbf{X}$ into P_2?

For the report: Answers to (***a***), (***b***) and (***c***).

Exercise 2. If $P_1 = [I\ 0]$ and

$$P_2 = \begin{pmatrix} 1 & 1 & 1 & 2 \\ 0 & 2 & 0 & 2 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \tag{2}$$

a) Compute the epipoles, by projecting the camera centers.

b) Compute the fundamental matrix, its determinant and verify that $e_2^T F = 0$ and $Fe_1 = 0$.

For the report: Complete solutions for (***a***) and (***b***).

Exercise 3. (OPTIONAL) Consider a general camera pair $P_1 = [I\ 0]$ and $P_2 = [A\ t]$.

a) Compute the epipoles, by projecting the camera centers. (You may assume that A is invertible.)

b) Show that for the fundamental matrix $F = [t]_\times A$ the epipoles will always fulfill

$$e_2^T F = 0 \quad \text{and} \quad Fe_1 = 0. \tag{3}$$

c) Given the above result explain why the fundamental matrix has to have determinant 0.

For the report: Optional, but if you want bonus points then report solutions for (***a***), (***b***) and (***c***).

Exercise 4. When computing the fundamental matrix F using the 8-point algorithm it is recommended to use normalization. Suppose the image points have been normalized using

$$\tilde{\mathbf{x}}_1 \sim N_1 \mathbf{x}_1 \text{ and } \tilde{\mathbf{x}}_2 \sim N_2 \mathbf{x}_2. \tag{4}$$

If $\tilde{F}$ fulfills $\tilde{\mathbf{x}}_2^T \tilde{F} \tilde{\mathbf{x}}_1 = 0$ what is the fundamental matrix F that fulfills $\mathbf{x}_2^T F \mathbf{x}_1 = 0$ for the original (un-normalized) points?

For the report: Answer is enough.

Exercise 5. In this exercise you will compute the fundamental matrix for the two images in Figure 1 of a part of the fort Kronan in Gothenburg.

Figure 1: `A3_ex5_kronan1.jpg` and `A3_ex5_kronan2.jpg`.

The file `A3_ex5_data.npz` contains `x1` and `x2` with matched points for the two images.

a) First compute normalization matrices N_1 and N_2. These matrices should subtract the mean and re-scale using the standard deviation, as in Assignment 2. Normalize the image points of the two images with N_1 and N_2 respectively.

b) Set up the matrix M in the eight point algorithm (use all the points), and solve the homogeneous least squares system using SVD. Check that the minimum singular value and $||Mv||$ are both small.

c) Construct the normalized fundamental matrix from the solution v. Don't forget to make sure that $\det(\tilde{F}) = 0$ for your solution. Check that the epipolar constraints $\tilde{\mathbf{x}}_2^T \tilde{F} \tilde{\mathbf{x}}_1 = 0$ are roughly fulfilled.

d) Compute the un-normalized fundamental matrix F (using the formula from exercise 4) and the epipolar lines $l = F\mathbf{x}_1$. Pick 20 points in the second image at random and plot these in the same figure as the image. Also plot the corresponding epipolar lines in the same image using the function `utils.draw_hom_line`. Are they close to each other?

e) Compute the distance between all the points and their corresponding epipolar lines and plot these in a histogram with 100 bins. What is the mean distance?

f) Try what happens if you do everything without normalization (that is, set $N_1 = N_2 = I$ and run the code again). What is the mean distance in this case?

Submit the code and for the report:

- the plot of the epipolar lines from (***d***),
- the histograms and mean epipolar distances from (***e***) and (***f***),

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Hints:

# Build M matrix for epipolar constraints
M = np.zeros((x1n.shape[1], 9))
for i in range(x1n.shape[1]):
    xx = np.outer(x2n[:, i], x1n[:, i])      # 3×3 outer product
    M[i, :] = xx.ravel()

# Reshape solution vector v to a 3x3 matrix
F = v.reshape(3, 3)

# Epipolar constraints, should be roughly zero
epi = np.sum(x2n * (Fn @ x1n), axis=0)

# Index vector with 20 random indices
ind = np.random.choice(x2.shape[1], size=20, replace=False)


# Epipolar lines
l = F @ x1
# Normalization factor (1×N)
norms = np.sqrt(l[0, :]**2 + l[1, :]**2)
# Normalize each line so first two components form unit normal
l = l / norms
# Compute distances and plot histogram
distances = np.abs(np.sum(l * x2, axis=0))
plt.hist(distances, bins=100)
plt.show()
```

Exercise 6. Consider the fundamental matrix

$$F = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

a) Project the 3D points

$$\mathbf{X}_1 = (1, 2, 3), \quad \mathbf{X}_2 = (3, 2, 1) \quad \text{and} \quad \mathbf{X}_3 = (1, 0, 1) \tag{5}$$

into the cameras $P_1 = [I\ 0]$ and $P_2 = [[e_2]_\times F\ e_2]$, where e_2 is the left epipole of F.

b) Verify that the projections satisfy the epipolar constraint $x_2^T F x_1 = 0$.

c) What is the camera center of P_2?

For the report: Complete solution for (***a***), (***b***) and (***c***).

Exercise 7.

a) Use the fundamental matrix F that you obtained in Exercise 5 to compute the camera matrices, using the method from Exercise 6.

b) Triangulate (with DLT) the 3D points using the two cameras.

Don't forget to normalize when triangulating. Note that since the point sets are the same as in Computer Exercise 5 the normalization matrices will also be the same. (Alternatively one could compute cameras and 3D points from the fundamental matrix $\tilde{F}$ obtained with the normalized points and transform the cameras afterwards. This also gives a valid solution, but it is a different one.)

c) Plot both the image, the image points, and the projected 3D points in the same figure.

d) Plot the 3D-points in a 3D plot. Does it look like the 3D scene?
(Hint: Recall Exercise 1 in Assignment 2...)

Submit the code and for the report:

- the 2D plot (with 2D points and projections overlaid on the image) from (***c***)
- the 3D plot (with the 3D points) from (***d***)

```
Hints:

from scipy.linalg import null_space

# Compute second epipole as the left nullspace of F
e2 = null_space(F.T)[:, 0]

# Constructs the cross-product matrix
e2x = np.array([
    [0,      -e2[2],  e2[1]],
    [e2[2],   0,     -e2[0]],
    [-e2[1],  e2[0],  0     ]
])
```

3 The Essential Matrix

Exercise 8. (**OPTIONAL**.) The goal of this exercise is to show that an essential matrix has two nonzero identical singular values.

Suppose the 3×3 skew symmetric matrix $[t]_\times$ has a singular value decomposition

$$[t]_\times = USV^T, \tag{6}$$

where U, V are orthogonal and S diagonal with non-negative elements.

a) Show that the eigenvalues of $[t]_\times^T[t]_\times$ are the squared singular values. (Hint: Show that $S^TS = S^2$ is the diagonalization of $[t]_\times^T[t]_\times$, see your linear algebra book.)

b) Verify that the eigenvectors of $[t]_\times^T[t]_\times$ fulfill

$$-t \times (t \times w) = \lambda w. \tag{7}$$

The cross-product of three vectors satisfy the following identity

$$u \times (v \times w) = (u \cdot w)v - (u \cdot v)w. \tag{8}$$

c) Show that $w = t$ is an eigenvector to $[t]_\times^T[t]_\times$ with eigenvalue 0 and that any w that is perpendicular to t is an eigenvector with eigenvalue $||t||^2$. Are these all of the eigenvectors?

d) Show that the singular values of $[t]_\times$ are $0, ||t||$ and $||t||$.

e) If $E = [t]_\times R$ and $[t]_\times$ has the SVD in (6), write an SVD of E. What are the singular values of E?

For the report: Complete solution if you want bonus, otherwise nothing.

Exercise 9. The file `A3_ex9_data.npz` contains the calibration matrix K for the two images in Computer Exercise 5.

a) Normalize the image points using the inverse of K.

b) Set up the matrix M in the eight point algorithm, and solve the homogeneous least squares system using SVD. Check that the minimum singular value and Mv are both small.

c) Construct the Essential matrix from the solution v. Don't forget to make sure that E has two equal singular values and the third one zero. Check that the epipolar constraints $\tilde{\mathbf{x}}_2^T E \tilde{\mathbf{x}}_1 = 0$ are roughly fulfilled.

d) Similar to Exercise 5, compute the fundamental matrix for the un-normalized coordinate system (this time using the K matrices instead of N), using your estimated Essential matrix.

e) Compute the epipolar lines $l = F\mathbf{x}_1$. Pick 20 of the detected points in the second image at random and plot these in the same figure as the image. Also plot the corresponding epipolar lines in the same figure using the function `utils.draw_hom_line`.

f) Compute the distance between the points and their corresponding epipolar lines and plot these in a histogram with 100 bins. What is the mean distance? How does this result compare to the corresponding result Exercise 5?

Submit the code and for the report:

- the plot of the epipolar lines from (***e***),
- the histograms and mean epipolar distances from (***f***),

Exercise 10. An essential matrix has the singular value decomposition

$$E = U\mathrm{diag}([1\ 1\ 0])V^T \tag{9}$$

where

$$U = \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \text{ and } V = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \tag{10}$$

a) Verify that $\det(UV^T) = 1$.

b) Compute the essential matrix and verify that $x_1 = (0, 0)$ (in camera 1) and $x_2 = (1, 1)$ (in camera 2) is a plausible correspondence.

c) If x_1 is the projection of $\mathbf{X}$ in $P_1 = [I\ 0]$ show that $\mathbf{X}$ must be one of the points

$$\mathbf{X}(s) = \begin{pmatrix} 0 \\ 0 \\ 1 \\ s \end{pmatrix}. \tag{11}$$

d) For each of the four solutions for P_2:

$$P_{2a} = [UWV^T \ u_3], \quad P_{2b} = [UWV^T \ -u_3] \tag{12}$$

$$P_{2c} = [UW^TV^T \ u_3], \quad P_{2d} = [UW^TV^T \ -u_3], \tag{13}$$

where

$$W = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{14}$$

and u_3 is the third column of U, compute s such that $\mathbf{X}(s)$ projects to x_2.

e) For which of the camera pairs is the 3D point $\mathbf{X}(s)$ in front of **both** cameras?

For the report: Complete solution.

Exercise 11.

a) For the essential matrix obtained in Computer Exercise 9 compute four camera solutions in (13). Remember to ensure that $\det(UV^T) = 1$ (see `Hints` below).

b) Triangulate the points using DLT for each of the four camera solutions, and determine for which of the solutions the points are in front of the cameras.

Since there is noise involved it might not be possible to find a solution with all points in front of the cameras. In that case select the one with the highest number of points in front of the cameras. (Note that the SVD of E is not unique. Different solutions may result in permutations of the order of the solutions for P_2. Therefore the code has to determine the number of points in front of the cameras to find the right solution.)

c) Compute the corresponding camera matrices for the original (un-normalized) coordinate system and plot the image, the points, and the projected 3D-points in the same figure. Do the errors look small?

d) Plot the 3D points and camera centers and principal axes in a 3D plot. Does it look like more reasonable compared to the result you got in Exercise 7?

For the report: Submit the code and plots from (***c***) and (***d***).

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Hints:

# Note that this returns the V matrix already transposed!
u, s, vt = np.linalg.svd(E)
if np.linalg.det(u @ vt) < 0:
    vt = -vt
E = u @ np.diag([1, 1, 0]) @ vt
# Note: Re-computing svd on E may still give U and V that does not fulfill
# det(U*V') = 1 since the svd is not unique.
# So don't recompute the svd after this step.
```